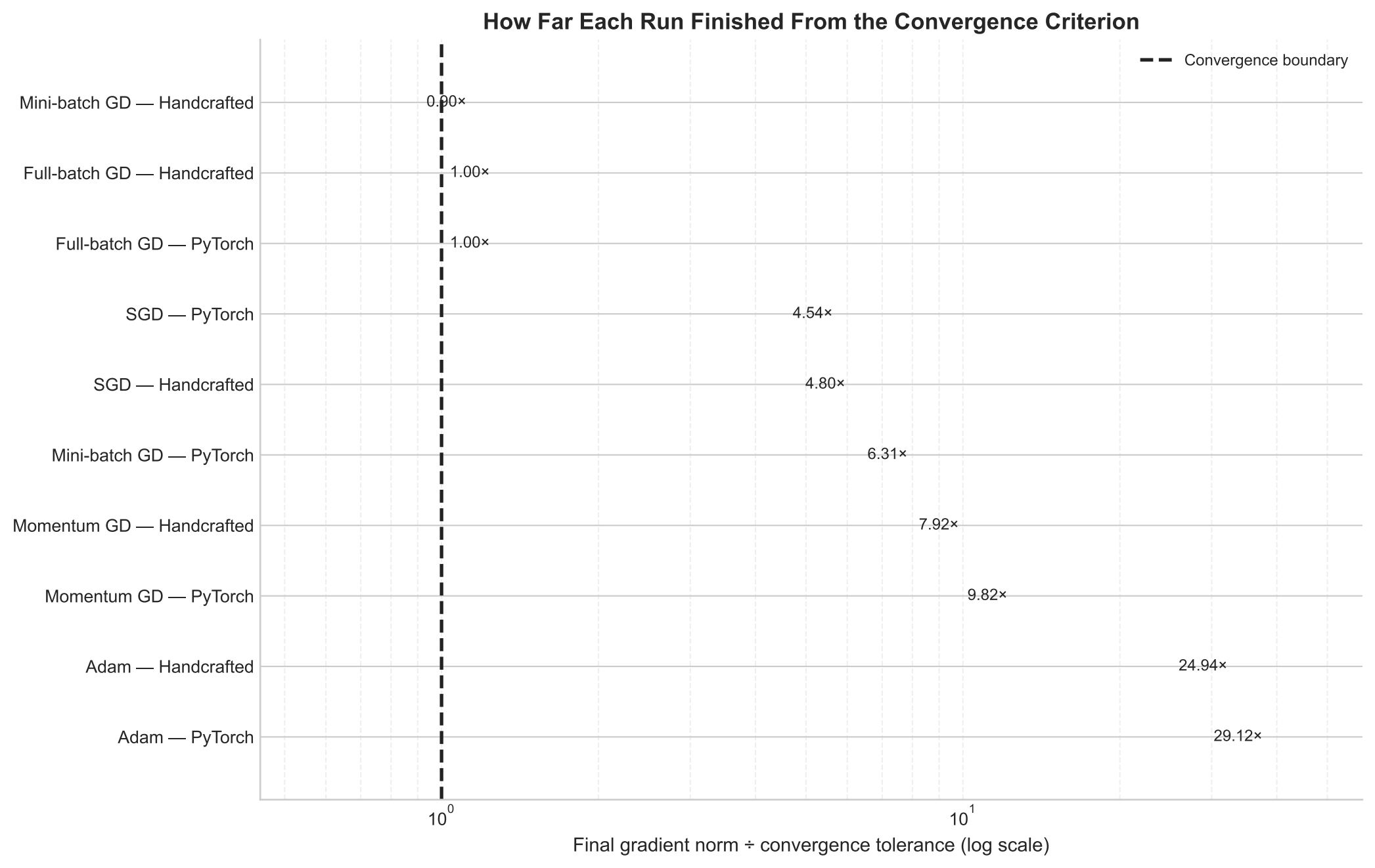
How Far Each Run Finished From the Convergence Criterion
Convergence boundary
Mini-batch GD — Handcrafted
Full-batch GD — Handcrafted
Full-batch GD — PyTorch
SGD — PyTorch
SGD — Handcrafted
Mini-batch GD — PyTorch
Momentum GD — Handcrafted
Momentum GD — PyTorch
Adam — Handcrafted
Adam — PyTorch
0.90×
1.00×
1.00×
4.54×
4.80×
6.31×
7.92×
9.82×
24.94×
29.12×
10^0
10^1
Final gradient norm ÷ convergence tolerance (log scale)